

Problem Set 3

Operations Research

Give rigorous proofs. Results proved in earlier problems may be used in later ones.

Problem 1

Let $A \in \mathbb{R}^{m \times n}$ have rank m , let $b \in \mathbb{R}^m$, and let $c \in \mathbb{R}^n$. Suppose that

$$Q = \{x \in \mathbb{R}^n : Ax = b, x \geq 0\} \neq \emptyset \quad \text{and} \quad \sup_{x \in Q} c^T x < +\infty.$$

Prove that there exists a basic feasible solution $x^* \in Q$ such that

$$c^T x^* \geq c^T x \quad \text{for every } x \in Q.$$

Your proof must establish attainment of the optimum, not assume it.

Problem 2

Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $c \in \mathbb{R}^n$. Consider

$$(P) \quad \begin{array}{ll} \max & c^T x \\ \text{subject to} & Ax \leq b, \\ & x \geq 0. \end{array}$$

- (a) Derive the dual program (D) , justifying its objective, constraints, and variable domains.
- (b) Prove that the dual of (D) is (P) .

Problem 3

For the primal–dual pair $(P), (D)$ of Problem 2:

- (a) Prove that $c^T x \leq b^T y$ for every primal feasible x and dual feasible y .
- (b) Prove that if feasible x and y satisfy $c^T x = b^T y$, both are optimal.
- (c) Prove that if (P) is unbounded above, then (D) is infeasible. Prove the corresponding statement when (D) is unbounded below.

Problem 4

For the same pair $(P), (D)$, assume that (P) is feasible and its objective is bounded above on its feasible set. Prove that both programs attain their optima and that

$$\boxed{\max_{Ax \leq b, x \geq 0} c^T x = \min_{A^T y \geq c, y \geq 0} b^T y.}$$

In particular, establish the existence of a dual feasible solution attaining the primal optimal value.